Two power laws force commutativity

Herstein, 2.2.5

Let G be a group such that $(ab)^3 = a^3b^3$ and $(ab)^5 = a^5b^5$ for all $a, b \in G$. Prove that G is abelian.

Proof. Let $a, b \in G$ be arbitrary. We show that $ab = ba$. We write powers as repeated products, e.g. $z^2 = zz$, $z^3 = zzz$, and use associativity freely.

First, for all $x, y \in G$, we claim that

$$(yx)^2 = x^2y^2.$$

Indeed,

$$x(yx)^2y = (xy)^3 = x^3y^3 = x(x^2y^2)y,$$

where the first and last equalities are obtained by expanding the powers and using associativity, and the middle equality is the hypothesis for the exponent 3 applied to x and y. Cancelling x on the left and y on the right gives $(yx)^2 = x^2y^2$.

Similarly, for all $x, y \in G$,

$$(yx)^4 = x^4y^4.$$

Indeed,

$$x(yx)^4y = (xy)^5 = x^5y^5 = x(x^4y^4)y,$$

where again the first and last equalities are expansions of powers, and the middle equality is the hypothesis for the exponent 5. Cancelling x on the left and y on the right gives $(yx)^4 = x^4y^4$.

Using these two identities with $x = a$ and $y = b$, we obtain

$$(a^2b^2)^2 = ((ba)^2)^2 = (ba)^4 = a^4b^4.$$

We next show that a^2 and b^2 commute. We have

$$a^2(b^2a^2)b^2 = (a^2b^2)^2 = a^4b^4 = a^2(a^2b^2)b^2,$$

where the first and last equalities follow by expanding the powers and using associativity, and the middle equality is the identity just proved. Cancelling a^2 on the left and b^2 on the right yields

$$b^2a^2 = a^2b^2.$$

Consequently,

$$(ab)^2 = b^2a^2 = a^2b^2,$$

where the first equality is the square identity above with $x = b$ and $y = a$, and the second equality is the commutativity of a^2 and b^2 just proved.

Finally,

$$a(ba)b = (ab)^2 = a^2b^2 = a(ab)b,$$

by expansion of powers and the preceding identity. Cancelling a on the left and b on the right gives $ba = ab$, hence $ab = ba$.

Since $a, b \in G$ were arbitrary, G is abelian. □